

Building GIS Capability in the Counterpart Department

A team that loses delivery capacity to two resignations is a team where knowledge was never written down. The counterpart department has the same fault line: its lowest measured competency is documenting a reproducible workflow, which means the work lives in individuals, not in the organisation.

I am treating capability-building and documentation as one task, not two, and I am making that call as the person who will own the outcome, not just deliver the week—every design choice below moves knowledge out of people and into procedures the department keeps when someone leaves.

1. What the evidence actually tells me

Three findings, and three decisions that follow directly from them—this is the difference between a training plan and a capability plan: the second treats evidence as a constraint on design, not as background colour.

Finding	What it rules out	What it requires
Self-rating vs. tested score correlation = 0.11 (no relationship); mean score 36-57% depending on measure	Using stated demand or confidence to decide content—people cannot report what they don't know they lack	Content set by measured gaps only; every topic opens with a task people attempt cold, so belief meets ability early and privately
Confident staff meeting a task they can't do tend to disengage to protect face	A conventional Day-1 pre-test read aloud, scored publicly	Day 1 opens as private, no-scores-aloud calibration, framed as closing a shared gap, not exposing individuals
0 of 21 staff have QGIS/ArcGIS access	Running the course as scheduled if software isn't installed and tested first	Software installation becomes a precondition, not a parallel administrative task (Section 6)

The decision I want to be explicit about, because it is the one a training coordinator would not make and a capability coordinator must: if software access slips, I recommend delaying the course rather than spending the budget. Skills not practised within days of learning decay fast; a five-day course into a zero-software environment produces a certificate and nothing the

department can use. That is a resourcing call I would take to whoever owns the budget before Day 1, not a caveat buried in a risk log—a senior coordinator's job is to make that trade-off visible and defend it, not to let the course run and let the failure surface quietly at 90 days.

Why this is a coordination problem, not a training problem

A department where competence lives in two or three individuals has a bus-factor problem it does not know it has—the 0.11 correlation above means management cannot even rely on staff to flag who those individuals are. Treating this as a training gap fixes the wrong layer: it improves individuals without changing the organisation's exposure. Everything from here—the level-4 "Steward" tier, the peer-reproduction exercise, the workflow-log requirement on every real assignment—is aimed at the organisational layer, on the assumption that I am accountable for departmental continuity, not for a course-completion count.

2. Competency framework

Generic beginner/intermediate/advanced labels are useless here precisely because self-labels don't predict ability. Each level below is defined by behaviour a supervisor can observe and verify—not by time served, and not by self-report.

Level	Behaviour a supervisor can observe
1. Assisted	Follows a written procedure with guidance; imports data, logs actions, asks for help at the right points
2. Supported	Cleans data and runs a basic join unaided; applies documented workflows consistently; spots obvious errors
3. Independent	Chooses the right method, validates results, writes metadata, produces work another officer can repeat
4. Steward	Writes/updates SOPs, reviews colleagues' work, mentors, sets quality standards—strengthens the department's capability not just their own

Six domains (data acquisition, data management/documentation, analysis, cartography, QA/governance, knowledge sharing) are scored against this ladder; the full matrix is Annex A. Progression requires evidence from three separate pieces of work, not one good day—this is a governance choice as much as a pedagogical one: it stops a single strong submission from inflating a competency record that a manager will later rely on for a staffing decision.

3. The five-day course

One rule: build competence before complexity. A baseline of 36/100 means the job is to make people solid on the core workflow, not to expose them to techniques they cannot yet support in production. Hands-on-to-instruction ratio is 70:30—adults build technical skill by doing the task and getting immediate correction, not by watching slides, and every hour of instruction I authorise has to justify itself against that ratio.

Day	Focus	Outcome	Exercise
1	Foundations & data quality	Apply validation rules; correct attribute errors	Clean a flawed facility register
2	Reproducible workflows (weakest skill)	Document a full cleaning procedure; peer-reproduce it	Swap files, reproduce a colleague's result
3	Spatial analysis for decisions	Apply joins; interpret result for service delivery	Find wards under-served by facilities
4	Cartography & QA	Design a readable map; evaluate against a checklist	Produce and peer-review a coverage map
5	End-to-end departmental case	Work a real request unaided; write an evidence note	Raw data → map → one-page finding, assessed

Day 2 carries the most weight deliberately: workflow documentation is both the weakest measured skill and the one that determines whether the department survives staff turnover. Day 4 is placed after data work on purpose—a polished map built on uncleaned data is worse than no map, and staff need to internalise that ordering before they're trusted with cartography they already over-value relative to its inputs.

4. What I am deliberately not teaching

Left out	Reason
Spatial statistics	Meaningless without reliable data and interpretation skill first

Left out	Reason
Remote sensing	Needs stronger fundamentals and software the department doesn't have
Python / automation	Automating a workflow you can't yet do by hand only automates the mistakes
Web GIS / dashboards	Sequenced after desktop competence and clean data are established
Machine learning, drone mapping	Outside the department's routine work; not a current need

The same discipline I apply to stated demand applies here: depth in what the department actually uses beats a thin, impressive-looking tour of what it cannot yet sustain. Cutting this list is itself a coordinator decision—every one of these is something a stakeholder may ask for by name, and part of the role is holding the line on sequencing against that pressure.

5. Assessment: measuring capability, not confidence

Because confidence doesn't predict ability here, both pre- and post-course sittings use the identical six-task practical instrument (100 marks: import/CRS 10, clean/validate 25, log & metadata 20, analysis 20, cartography 15, interpretation note 10), scored 0-5 per task against a fixed rubric. The post-test uses a matched dataset—same size, same error categories, same difficulty, different values—so a gain reflects learning, not memorised answers. Full item wording and rubric: Annex C; a ready-to-deliver 90-minute session on the weakest domain (reproducible cleaning), designed so a colleague can run it without me in the room, is Annex B.

One design choice I want to flag as coordinator-level rather than trainer-level: results are reported as a distribution (lowest/median/highest), never a single mean. An average hides exactly the people a capability programme exists to catch. If I am accountable for whether this department can function without external support, I need to know who is still below the independent-work threshold at Day 5, not that the cohort average looks acceptable.

6. The 90 days after the course—where capability actually gets decided

The course is the smaller half of this plan. Capability counts once people do real departmental work to standard, unsupervised. Four things run in parallel, and I hold a specific owner accountable for each rather than leaving them as shared aspirations:

- Fix the software constraint first (before Day 1, non-negotiable). QGIS LTR - free, no licensing barrier—installed and tested on every workstation, a named IT contact, a shared data folder. If this slips, the course slips; I would rather move the start date than run the course into an environment that guarantees the skill decays within a week.
- Days 1-30 real work, not more exercises. Each officer completes ≥ 2 genuine departmental assignments using the standard templates; every submission arrives with its workflow log, metadata, QA check and supervisor sign-off. This is what converts a taught skill into a departmental habit rather than a course memory.
- Days 31-60 structured transfer. Fortnightly mentoring pairs, a monthly problem clinic, peer review against the same QA checklist used in training.
- Days 61-90 independent delivery + feedback loop. A short lessons-learned session updates the SOPs—the department's documentation improves because the course ran, not just the people.

Measure	Baseline	90-day target
Staff with working GIS software	0 / 21	21 / 21
Documentation competency	1.5 / 5	$\geq 4 / 5$
Demonstrated capability (re-test)	36 / 100	$\geq 75 / 100$
Real assignments delivered to standard	0	2 per officer
Outputs passing QA on first submission	not yet tracked	$\geq 80\%$
Staff able to mentor a colleague	0	$\geq 4 / 21$

Risk ownership, stated plainly rather than left generic

The three failure modes that quietly kill programmes like this are software slipping, management not protecting practice time, and staff drifting back to undocumented habits once the facilitator leaves. Each has a named counter built into the design above, not a mitigation note appended afterward: install-before-train removes the first; getting management to ring-fence practice time before the course starts (a conversation I would have at the budget-approval stage, not after Day

5) removes the second; making documented workflow and sign-off the normal way work is submitted—not an optional extra reviewed later—removes the third.

Accountability, named rather than shared. A plan with no named owner per line item is a wish list; I would not present this to my own management as one.

Risk / dependency	Owner	Escalation trigger
Software not installed/tested	Department IT contact, confirmed by me	Any workstation untested 5 working days before Day 1; I escalate to the department head directly
Practice time not ring-fenced	Line supervisors, agreed before Day 1	An officer misses their Day-30 assignment count; raised at the fortnightly mentoring check, not left to the 90-day review
Workflow log/sign-off skipped on real work	Supervisor sign-off, spot-checked by me	Any submission without a log is returned unsubmitted, no exception
Cohort member still below Level 2 at Day 90	Named mentor pair	Individual coaching plan, not folded into the average and forgotten

Closing: Beyond 90 days

The 90-day plan closes with an SOP update, not a handover to nobody—the Level-4 Stewards identified during the course become the standing owners of those SOPs, and the department's own QA checklist (not an external consultant) becomes the mechanism that keeps standards from drifting once this engagement ends. That is the actual exit criterion: not that I have delivered a course, but that the department no longer needs me to keep its own workflow alive.

The department's real problem is not that people cannot make maps. It is that the work is undocumented and lives in individuals, which is exactly what makes a team fragile when someone leaves—the same fragility a two-resignation crisis exposes anywhere. Every part of this design pushes one direction: get the tools in place first, teach the core workflow well, write everything down in a form someone else can repeat, and prove the skill on real work within

ninety days. That last test—not the certificate, not the course-week satisfaction score—is the only one I would stand behind in front of the department's own management.

Annex A: Competency framework in full

Six domains, each scored against the four-level ladder in Section 2. Moving up a level requires evidence from at least three separate pieces of work.

Domain		Covers			Why it matters	
Data acquisition & validation		Collecting and checking spatial/attribute data			Reliable inputs—everything downstream depends on it	
Data management & documentation		Cleaning, organising, versioning, documenting			Reproducibility and institutional memory	
GIS analysis		Applying the right spatial method			Turns data into evidence for decisions	
Cartography & communication		Clear, accurate maps and findings			Makes the work usable by managers	
QA and governance		QA/QC, metadata standards, SOPs			Consistency, transparency, auditability	
Collaboration & knowledge sharing		Documenting, reviewing, mentoring			Resilience—reduces dependence on individuals	

Competency Progression Matrix Across Levels

Domain	L1 Assisted	L2 Supported	L3 Independent	L4 Steward
Data acquisition	Imports via SOP	Validates unaided	Designs the checks	Sets dept. standards
Data cleaning	Cleans with guidance	Cleans independently	Optimises the workflow	Improves org. workflow
Analysis	Runs a guided analysis	Picks the right tool	Produces decision-ready analysis	Reviews others' methods
Cartography	Produces a basic map	Produces a finished map	Sets mapping standards	Advises on visualisation
QA and metadata	Completes a checklist	Applies QA consistently	Updates standards	Reviews others' work
Knowledge sharing	Documents own	Shares	Mentors	Leads the

Domain	L1 Assisted	L2 Supported	L3 Independent	L4 Steward
	work	workflows	colleagues	practice group

Annex B: Ninety-minute session, ready for someone else to deliver

Session: Building a reproducible data-cleaning workflow | Duration: 90 min | Domain: Data management and documentation | Target: Level 2, Supported Practitioner

The baseline found documentation the weakest skill (1.5/5). This session fixes the department's highest organisational risk—cleaning work nobody else can repeat. By the end, a participant has both cleaned a dataset and documented it well enough for a colleague to reproduce the exact result from the write-up alone.

- Learning outcomes:

- Explain in one sentence why a repeatable workflow matters more than a fast one;
- Clean a health-facility dataset using an agreed sequence;
- Record every step in the standard workflow log and metadata sheet;
- Reproduce a colleague's result from their documentation alone and give QA feedback.

Materials: Per participant: laptop with QGIS, exercise dataset, workflow log template, metadata template, QA checklist, USB backup. Facilitator: projector, demonstration machine, this guide.

Dataset: "District Health Facilities"—250 facility records, 18 ward boundaries, 460 road segments, 1,200 settlement points, EPSG:4326. Planted defects mirror real partner-supplied data: duplicate facility IDs, spelling variants of the same facility, missing coordinates, null attributes, invalid facility types, inconsistent capitalisation, a few wrong ward codes.

Session Agenda

Time	Stage	Facilitator action
0-10 min	Why we're here	Ask: "If another officer had to update this in six months, could they, without you?" Documentation is organisational memory, not paperwork.
10-20 min	Demonstration	One cycle only: open data, spot an error, fix it, log it immediately. Do not clean the whole file for them.
20-60 min	They do it	Participants clean and log as they go. Circulate constantly; help only when stuck; never take the keyboard. Prompt with "how did you decide that?", not "that's wrong."
60-75 min	Peer reproduction	Swap machines; reproduce a colleague's output from

Time	Stage	Facilitator action
		documentation alone; note exactly where it breaks.
75-90 min	Debrief	Which errors needed judgement; where documentation fell short; capture recurring errors for later coaching.

Exercise brief to participants: "The M&E unit has received a facility dataset combined from several implementing partners. Before it can be used for quarterly reporting it must be cleaned, validated and documented so the result can be reproduced. Import the data into QGIS; work on a copy, never the original; find and fix the data-quality problems; remove duplicates; standardise facility names; fill missing values where you reliably can; save the cleaned layer; complete the workflow log and metadata sheet as you go. You have 40 minutes."

Model answer: Import and confirm CRS; make a working copy before editing; remove duplicate facility IDs; standardise spelling/capitalisation against an agreed name list; correct invalid facility types; fill missing ward codes from the boundary layer where possible and flag the rest; validate coordinates against ward boundaries; log each action with a short reason; complete metadata. Order can vary—marks are for a reproducible result, not one fixed sequence.

Facilitator Guidance & Common Errors

Error expected	Facilitator response
Editing the original file directly	Stop; show how to make a working copy first; explain why the source stays untouched
Cleaning everything, then writing up at the end	Redirect to logging each step as it happens
Metadata left half-empty	Point back to the checklist; ask for the missing fields
Deleting duplicates with no reason recorded	Ask them to justify each removal in the log—"looked wrong" is not a reason
Inconsistent naming decisions	Introduce the standard name list

Session Assessment: Scoring (100 total): Identifies errors (20) | Cleans accurately (20) | Workflow log complete (20) | Metadata complete (15) | Result reproducible by a peer (20) | Takes part in peer review (5).

Annex C: Pre- and post-course instrument

Both sittings are identical in structure, marked with the same rubric. The post-test uses matched dataset "B" (same record count, error categories and difficulty as "A", different values) so gains reflect learning, not memory. Re-run at 90 days to test whether capability held once people were back on real work.

Item	Task	Marks
1	Import three layers and confirm the coordinate system	10
2	Identify and correct duplicates, nulls, inconsistent names, wrong categories; save the cleaned layer	25
3	Complete the workflow log, metadata sheet and QA checklist	20
4	Determine which facilities fall outside a stated service distance (distance and source given, not asserted)	20
5	Produce a finished map: title, legend, scale, north arrow, sensible symbology	15
6	Write a one-page interpretation of the result for a manager	10

Scoring Rubric per Item (0-5, weighted to item marks)

Score	Meaning
0	Not demonstrated / incomplete
1	Major errors, needs substantial help
2	Partial, correct only with significant guidance
3	Correct with minor errors
4	Correct and independent
5	Correct, independent, well documented and quality-assured

Results are reported as a distribution (lowest, median, highest, share reaching each band) across the 21 staff, never a single mean, so weak performers stay visible.

Score Band	Interpretation
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Score Band	Interpretation
0 - 39	Needs close supervision
40 - 59	Handles routine work with help
60 - 79	Works independently
80 - 100	Can mentor others